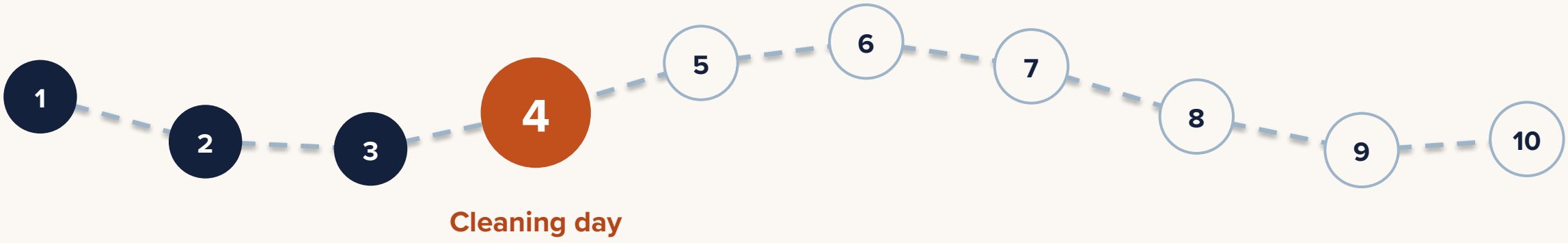
PYTHON FOR SUPPLY CHAIN

Session 4 · Thu 26 Nov 2026

Cleaning day

pandas, missing values, filters, new columns



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Today

1

Read and look

`read_csv` `head` `info`

2

Missing values

`isna` `sum`

3

Filters

`df[mask]` `&` `|` `~`

4

Cleaning rules

`6 rules, 1 log`

5

New columns

`revenue` `month`

6

Sort, count, save

`value_counts` `to_csv`

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pandas and the DataFrame

column names

	invoice	quantity	price	country
0	536365	6	2.55	United Kingdom
1	536365	6	3.39	United Kingdom
2	536370	24	3.75	France
3	536389	6	2.10	Australia

index

one Series

DataFrame ≈ a sheet

Series ≈ a column

index ≈ row number, from 0

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read_csv opens a file

`/content/drive/MyDrive/`	`Python for Supply Chain/`	`data/`	`online_retail_ii_lesson.csv`
your Drive	course folder	folder	file

```
folder = "/content/drive/MyDrive/Python for Supply Chain"
sales = pd.read_csv(folder + "/data/online_retail_ii_lesson.csv")
```

One wrong letter: FileNotFoundError

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Look first

`.head()`
first rows

`.shape`
rows, columns

`.info()`
types, filled cells

`.describe()`
count, mean, min, max

```
sales.shape
→ (105335, 8)
```

no brackets: a fact brackets: an action

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One column, one type

int64

`6  24  3906`

Quantity

float64

`2.55  17850.0`

Price, Customer ID

object

`"LANTERN"`

Invoice, Description, Country

datetime64

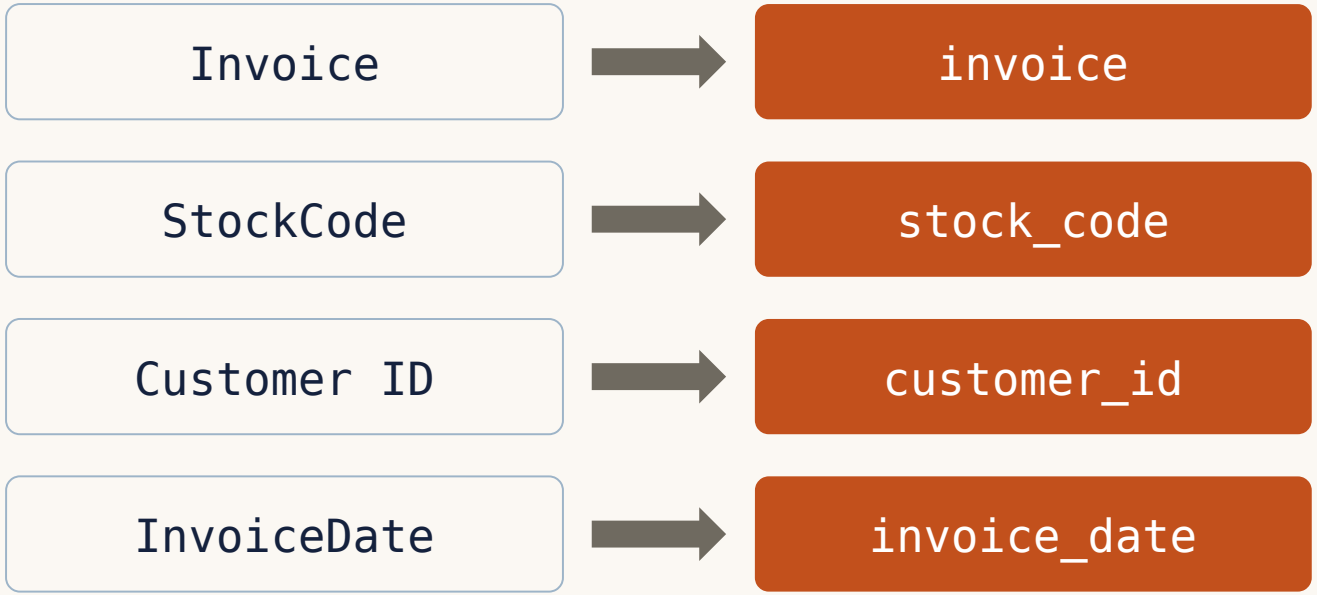
`2010-12-07 10:15`

after pd.to_datetime

After read_csv, dates are text. Convert them first.

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rename gives columns new labels

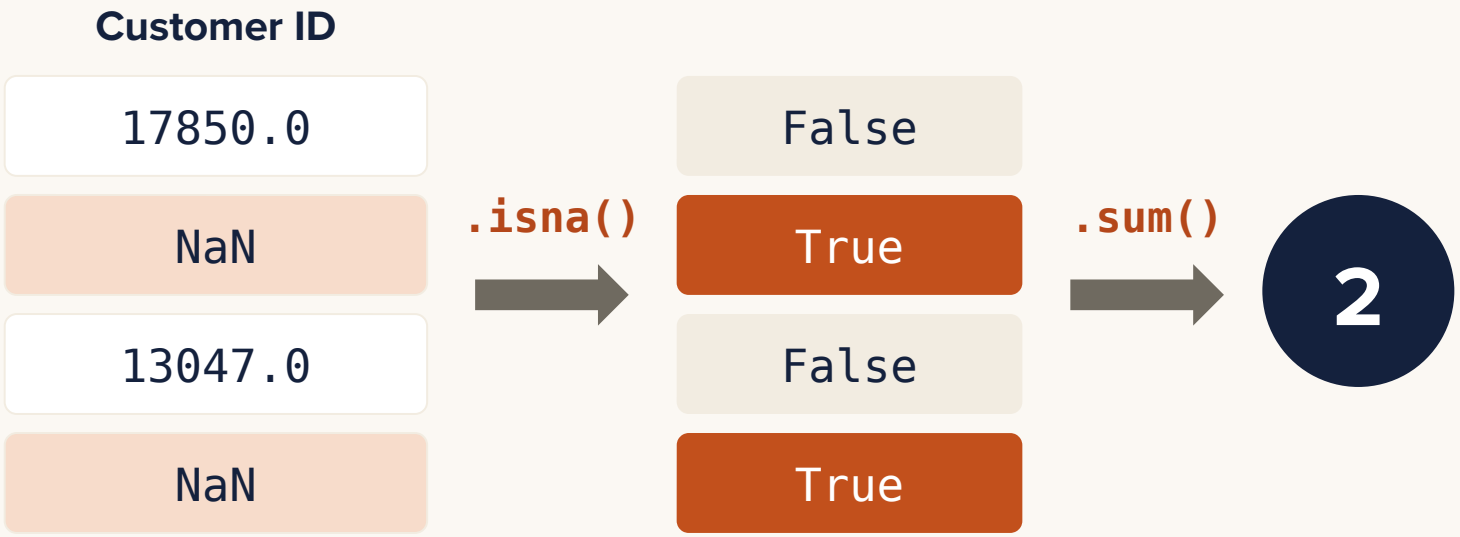



```
names = {
    "Invoice": "invoice",
    "Customer ID": "customer_id",
}
sales = sales.rename(columns=names)
```

After rename, sales["Country"] gives KeyError

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NaN: an empty cell




```
sales.isna().sum()
```

```
→ ...
→ Description    304
→ Customer ID  36210
→ ...
```

True = 1, False = 0

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Keep, drop or fill?

today

Keep

`(nothing)`

the question does not need it

Drop

`dropna(subset=[...])`

the question needs it

Fill

`fillna(0)`

a safe default exists

The question decides.

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A condition asks every row

United Kingdom		False
Germany	== "Germany"	True
France		False
Germany		True
EIRE		False



== equal	!= not equal
> bigger	>= bigger or equal
< smaller	<= smaller or equal

```
is_germany = sales["country"] == "Germany"
```

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df[mask] keeps the True rows

United Kingdom	False
Germany	True
France	False
Germany	True
EIRE	False



germany

Germany
Germany

sales stays the same

```
germany = sales[is_germany]
germany.shape
```

```
→ (1741, 8)
```

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Two conditions: & | ~

&

both true

```
(a > 0) & (b > 0)
```

|

at least one

```
(a > 0) | (b > 0)
```

~

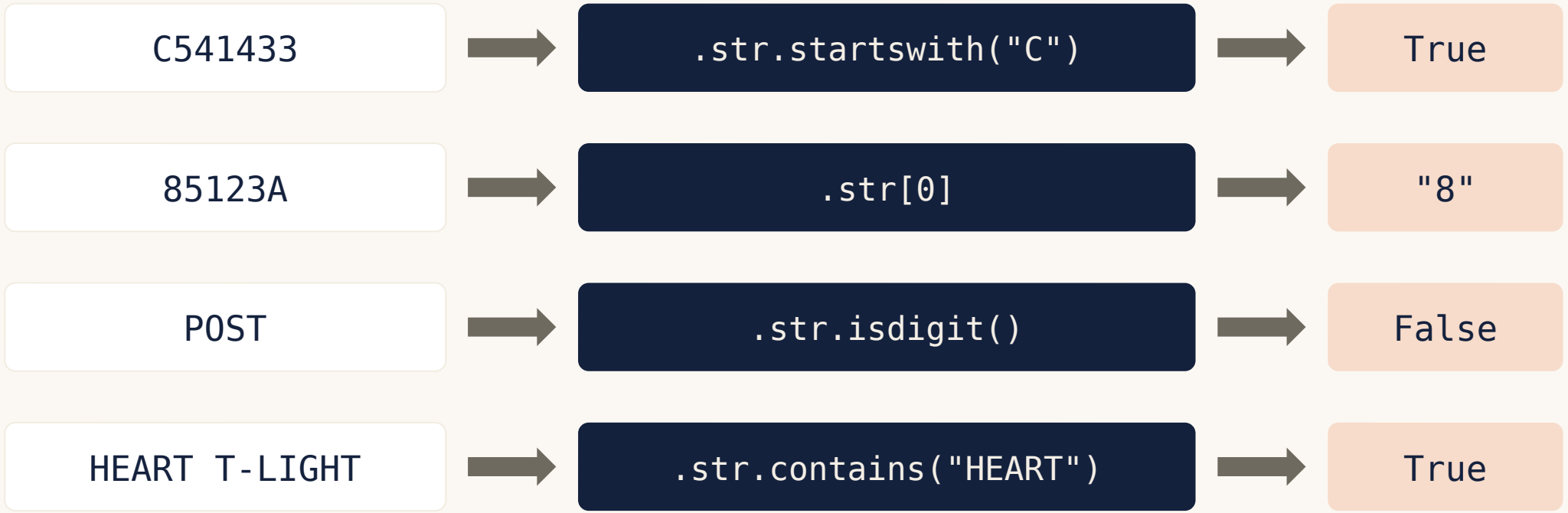
flip True/False

```
~is_germany
```

Words and / or → ValueError. Use symbols and brackets.

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.str asks questions about text



Text columns only. Tools can be chained.

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Six cleaning rules

1

Duplicates

same row twice

2

Cancelled

invoice starts with C

3

Quantity 0 or less

returns, stock fixes

4

Price 0 or less

free items, adjustments

5

Not a product

POST, DOT, M

6

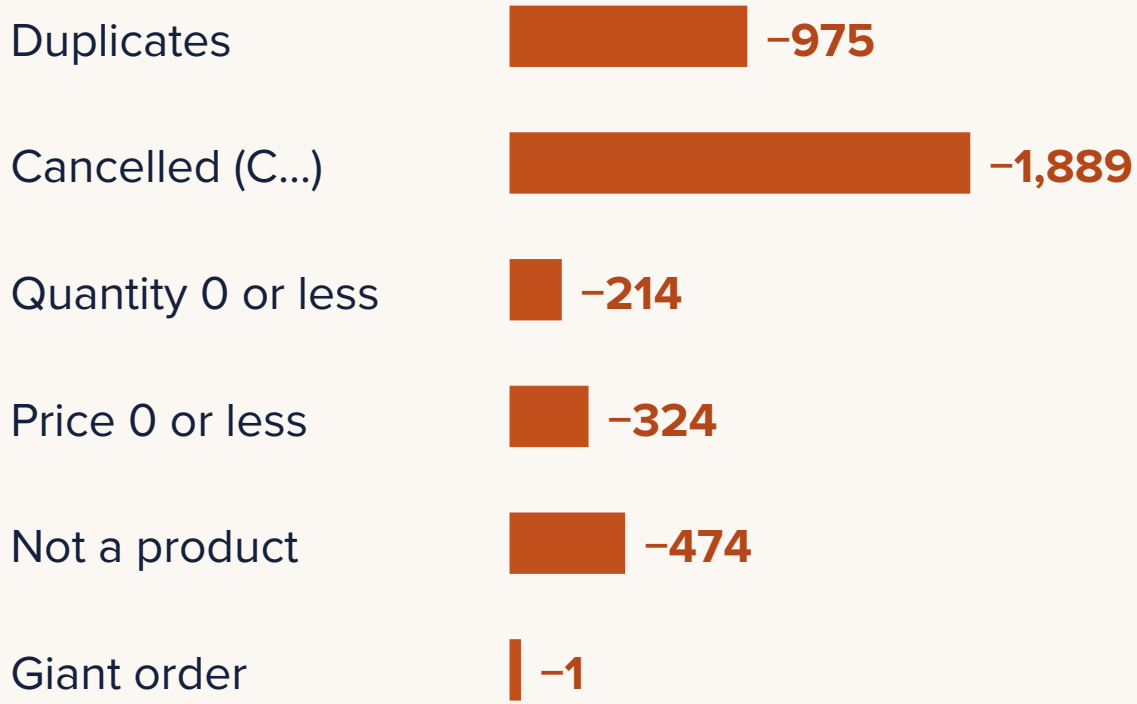
Giant order

cancelled 16 minutes later

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The cleaning log

105,335
raw rows



101,458
clean rows

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A filter makes a new table




```
sales_clean = sales_clean[mask]
sales_clean = sales_clean.copy()
```

Run cells from top to bottom.

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A new column, every row at once

```
df["revenue"] = df["quantity"] * df["price"]
```

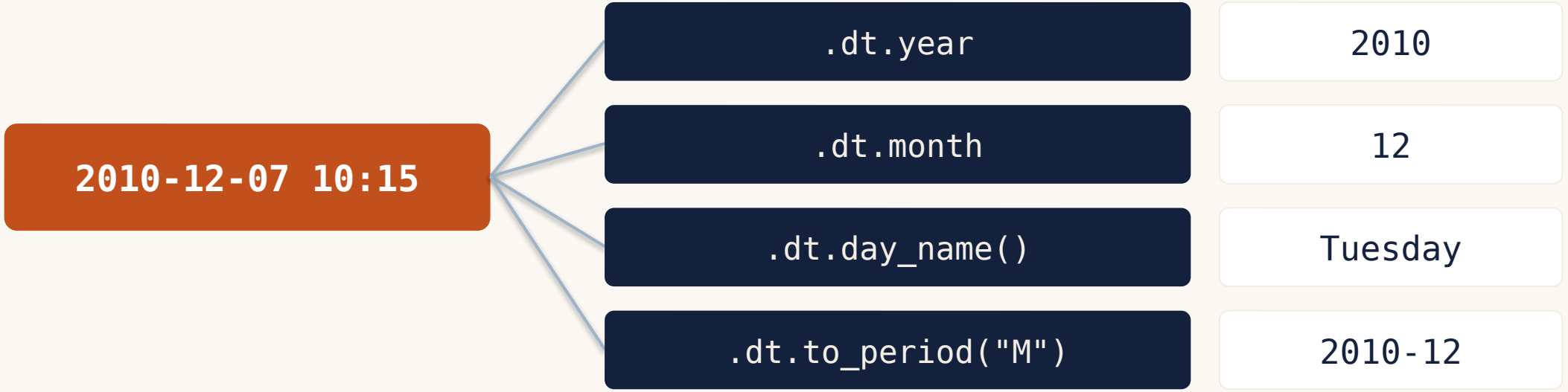
quantity		price		revenue
6		2.55		15.30
6	×	3.39	=	20.34
24		3.75		90.00

No drag-down. No auto-update.

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Text becomes a real date

```
sales["invoice_date"] = pd.to_datetime(sales["invoice_date"])
```

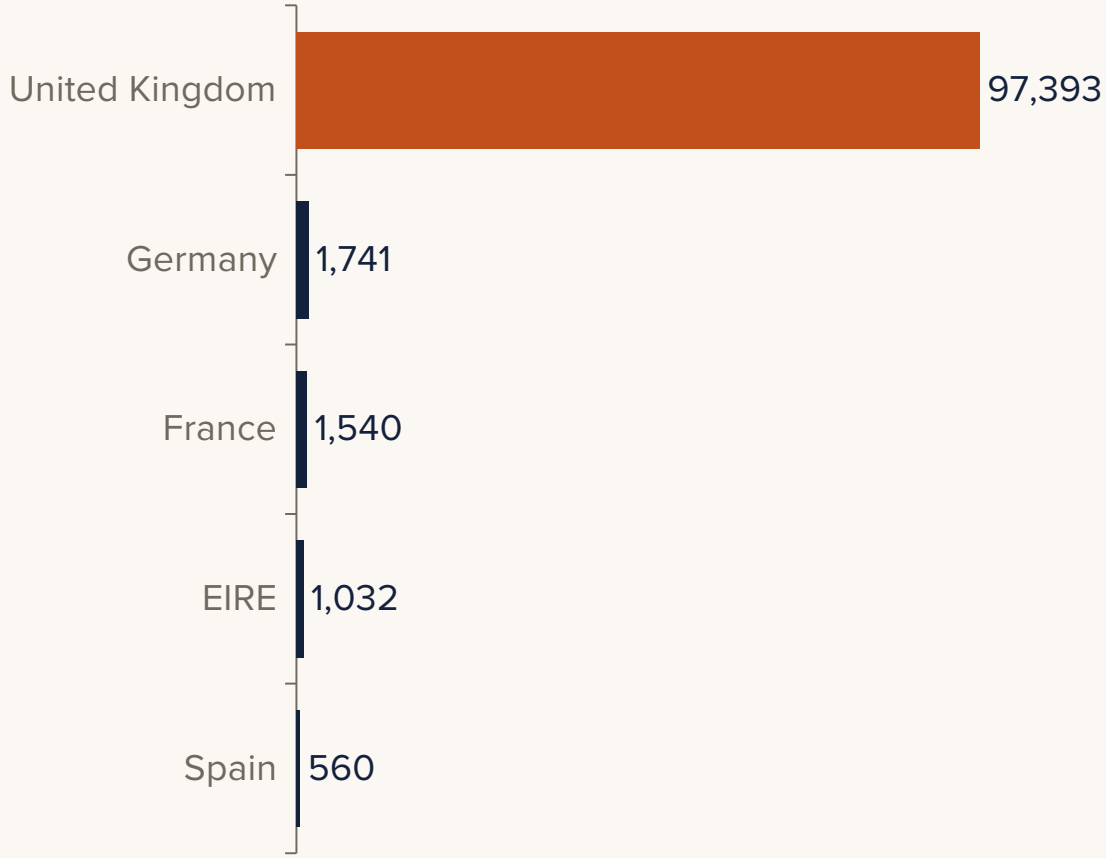


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Count rows per value

```
sales["country"].value_counts()
```

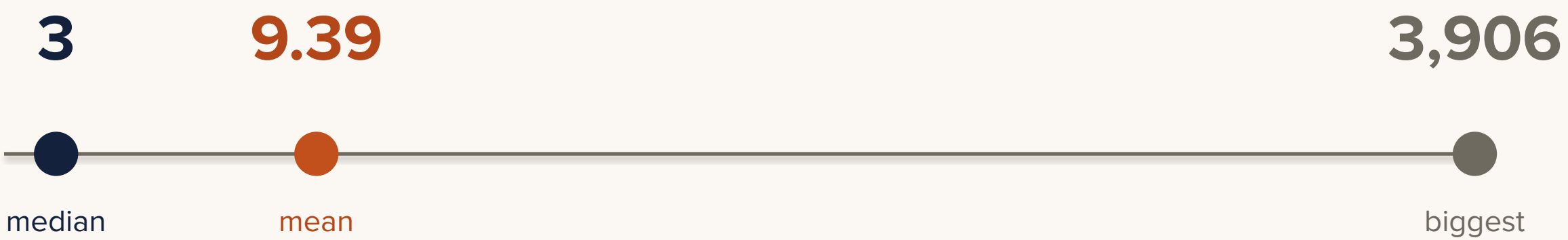
`.value_counts()`	rows per value
`.sort_values()`	biggest first
`.nunique()`	how many different



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Mean or median?

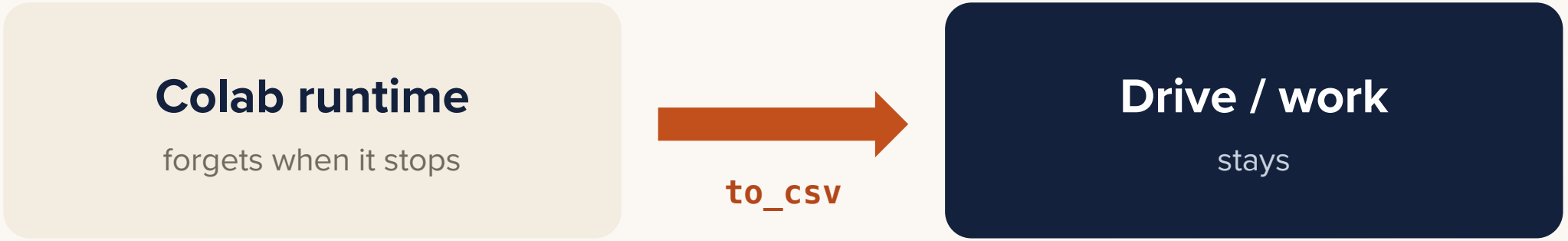
Quantity per row, clean data



`.sum()`	`.mean()`	`.median()`	`round(x, 2)`
total	average	middle	2 decimals

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Save to Drive, or lose it




```
sales_clean.to_csv(folder + "/work/sales_clean.csv", index=False)
```

index=False: no extra column of row labels

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The giant order

74,215 jars × **£1.04 each** = **£77,183.60**

541431	+74,215 jars	18 Jan 2011, 10:01
C541433	−74,215 jars	18 Jan 2011, 10:17

Same customer, same product, 16 minutes later. The sale never happened.

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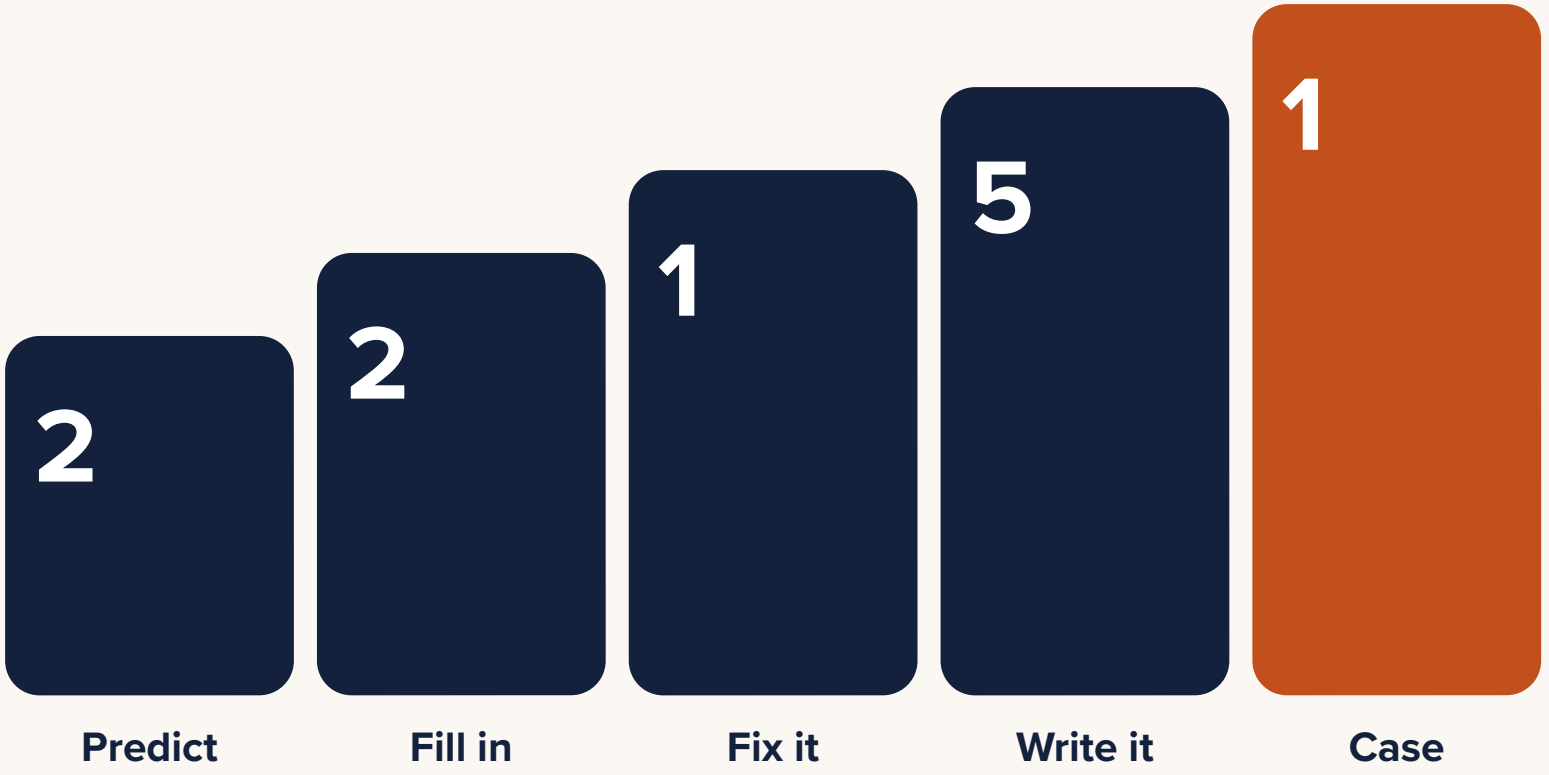
Your cleaning toolkit

Code	Does
`pd.read_csv(path)`	open a CSV file
`.head() .shape .info()`	look first
`.isna().sum()`	count empty cells
`.rename(columns={...})`	new column names
`df[mask]`	keep the True rows
`&  |  ~`	and, or, not
`.str...`	text tools

Code	Does
`.drop_duplicates()`	drop repeats
`.copy()`	Save As
`df["new"] = ...`	add a column
`pd.to_datetime() .dt`	real dates
`.sort_values() .value_counts()`	sort and count
`.sum() .mean() .median()`	summary numbers
`.to_csv(path, index=False)`	save to Drive

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Now in Colab



Notebook

s4_cleaning_day

Save

sales_clean.csv

Homework

2 extras

Recall

Fri 27 to Sun 29

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Thank you

Next: Session 5

Who are the star products?

Mon 30 Nov



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